Lab Mid

CSE 4308

DATABASE MANAGEMENT SYSTEMS LAB

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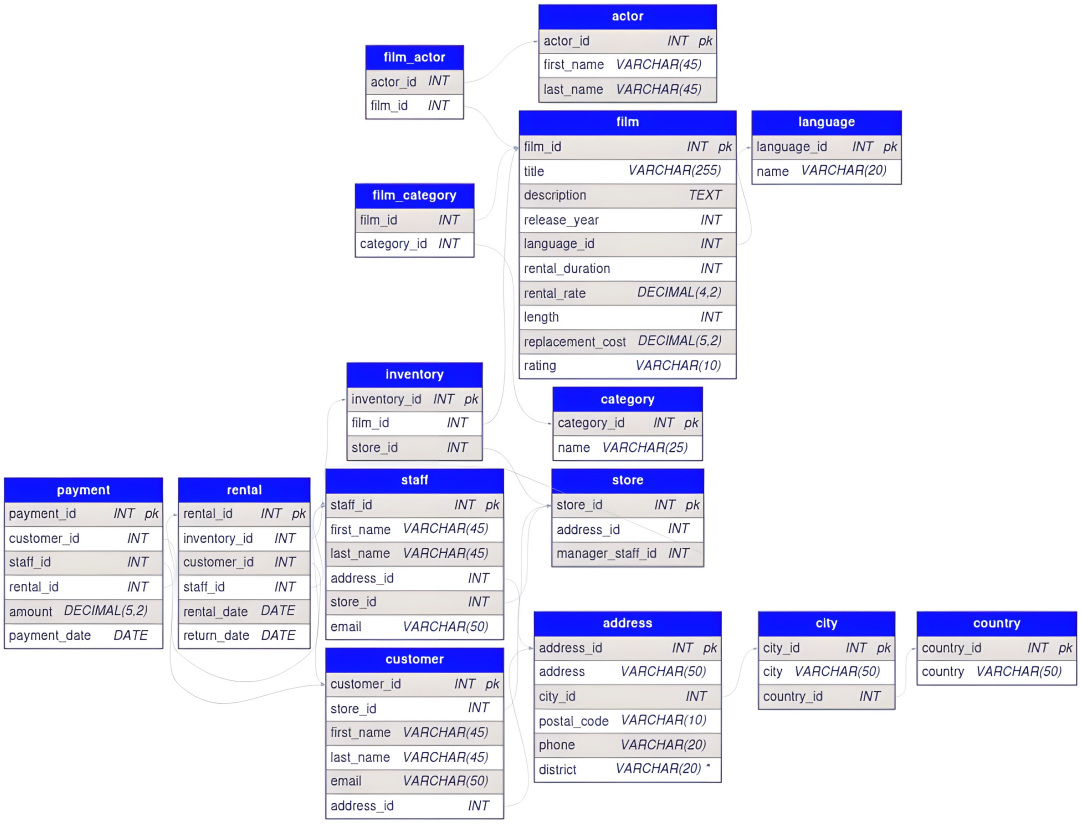
DBMS Lab Mid



1. Display the film title and release year for all films released after 2005.

```
SELECT title, release_year
FROM film
WHERE release_year > 2005;
```

2. For each district, show the number of customers living there. Only include districts with more than 5 customers.

```
SELECT a.district, COUNT(c.customer_id) AS customer_count
FROM customer c
JOIN address a ON c.address_id = a.address_id
GROUP BY a.district
HAVING COUNT(c.customer_id) > 5;
```

3. List all stores along with the total number of inventory items they have.

```
SELECT s.store_id, COUNT(i.inventory_id) AS total_inventory
FROM store s
LEFT JOIN inventory i ON s.store_id = i.store_id
GROUP BY s.store_id;
```

4. For each film rating (G, PG, PG-13, R), calculate the average rental rate.

```sql
SELECT rating, AVG(rental_rate) AS avg_rental_rate
FROM film
WHERE rating IN ('G', 'PG', 'PG-13', 'R')
GROUP BY rating
ORDER BY rating;
```

5. List rentals where the rental month and return month are the same.

```sql
SELECT rental_id, rental_date, return_date
FROM rental
WHERE EXTRACT(MONTH FROM rental_date) =
      EXTRACT(MONTH FROM return_date);
```

6. Find all customers whose first names start with 'A' and whose emails contain '.org'.

```sql
SELECT first_name, last_name, email
FROM customer
WHERE LOWER(first_name) LIKE 'a%'
  AND LOWER(email) LIKE '%.org%';
```

7. Display the customer ID and email for customers whose email length is greater than 20 characters.

```sql
SELECT customer_id, email
FROM customer
WHERE LENGTH(email) > 20;
```

8. Display each film title along with the total number of copies available in inventory.

```sql
SELECT f.title, COUNT(i.inventory_id) AS total_copies
FROM film f
JOIN inventory i ON f.film_id = i.film_id
GROUP BY f.film_id, f.title
ORDER BY total_copies DESC;
```

9. Find the actors who acted in both 'Comedy' and 'Action' films.

```sql
SELECT a.actor_id, a.first_name, a.last_name
FROM actor a
WHERE a.actor_id IN (
    SELECT actor_id
    FROM film_actor fa
    JOIN film_category fc ON fa.film_id = fc.film_id
    JOIN category c ON fc.category_id = c.category_id
    WHERE c.name = 'Comedy'
)
AND a.actor_id IN (
    SELECT actor_id
    FROM film_actor fa
    JOIN film_category fc ON fa.film_id = fc.film_id
    JOIN category c ON fc.category_id = c.category_id
    WHERE c.name = 'Action'
);
```

10. For each store, find the customer(s) who made the highest number of rentals in that store.

```sql
SELECT i.store_id, c.first_name, c.last_name, COUNT(*) AS rental_count
FROM inventory i
JOIN rental r ON i.inventory_id = r.inventory_id
JOIN customer c ON r.customer_id = c.customer_id
GROUP BY i.store_id, c.customer_id, c.first_name, c.last_name
HAVING COUNT(*) = (
    SELECT MAX(cnt)
    FROM (
        SELECT COUNT(*) AS cnt
        FROM inventory i2
        JOIN rental r2 ON i2.inventory_id = r2.inventory_id
        WHERE i2.store_id = i.store_id
        GROUP BY r2.customer_id
    ) x
);
```